

Samee Arif

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EDUCATION

Ph.D. in Computer Science and Engineering

University of Michigan - Ann Arbor (Advised by Prof. Rada Mihalcea)

Aug. 2025 - May 2030

GPA: 4.00/4.00

Bachelor of Science in Computer Science

Lahore University of Management Sciences

Sep. 2019 - May 2023

RESEARCH & TECHNICAL EXPERIENCE

Graduate Research Assistant | Language and Information Technologies (UMich)

May 2025 - Present

- Constructed a 75-country cultural reasoning benchmark of ~7.9K from Cultural Atlas, using a deontic-force taxonomy to route each norm to a valid question frame, with scenarios grounded in paired norm and Schwartz-value evidence.
- Analyzed country-conditioned value alignment using activation probing, representational similarity analysis, residual-stream steering, and SFT/GRPO ablations across country-specific and counterfactual training splits.

Research Associate | Center for Speech and Language Technologies (LUMS)

Jun. 2024 - Jan. 2025

- Developed a multi-model pipeline for synthetic DPO/KTO data generation across 6 LLMs and 4 benchmarks; fine-tuned models on the generated data, achieving 71.8%/73.8% win rates over Llama/Gemma baselines.
- Designed and deployed Kahaani, a multi-agent storytelling system spanning text, speech, music, and video; evaluated the full system in a human study with 3 parent-child pairs, scoring 4/5 child experience and 5/5 age-appropriateness.

Machine Learning and Software Engineer | ActualAlz

Aug. 2023 - Jun. 2024

- Engineered and deployed AI assistant for university applications and SOP writing, supporting 100+ global placements.

Research Assistant | LUMS

Aug. 2021 - May 2023

- Compared general and fine-tuned LMs across 14 Urdu NLP tasks; specialist models outperformed on 12/14 tasks.
- Proposed EATS, an answer-span-preserving translation method, and constructed an Urdu QA corpus of 124,745 pairs.
- Built Urdu conversational speech benchmark (471 recordings, 1.3 hours); fine-tuned and evaluated 9 ASR models.
- Built the first Urdu Newspaper Benchmark (9,982 sentences) and fine-tuned VLMs for OCR, reducing WER by 6.13%.

PROJECTS

Incremental Completion Attacks on Aligned Language Models

- Proposed a jailbreak framework that elicits harmful outputs by generating one word at a time toward a harmful objective, gradually shifting the model's internal safety state and bypassing alignment through context accumulation.
- Achieved 99.62% ASR on Vicuna-13B, 90.58% on DeepSeek-R1-Distilled, and 87.31% on Llama-3.1-8B; evaluated across 3 benchmarks and 21 models, with mechanistic probes revealing suppressed refusal/safety representations.

Developmentally-Aware Evaluation of Child-Facing LLMs

- Developed KIDBench, a developmental-psychology-grounded benchmark evaluating 13 LLMs across single- and multi-turn child interactions, revealing 10–30% gains from explicit age cues and 6–24% multi-turn degradation.
- Trained KIDGuardLlama, a child-safety evaluator, and KIDLlama, a child-oriented response model, on 22K+ child-safe responses using SFT + Critique-GRPO; human evaluators preferred KIDLlama 60% vs. 36% for the strongest baseline.

One-Shot Alignment on LLM Fairness Benchmarks

- Evaluated one-shot GRPO and ICL for fairness alignment, raising Qwen-2.5-7B BBQ accuracy from 79.9% to 92.9% and 99.0%, respectively; one-shot GRPO captured ~80% of the improvement achieved by large-scale RLHF.
- Traced benchmark gains to a simple “missing-evidence” reasoning strategy rather than broad fairness alignment using centered kernel alignment to compare internal representations, plus logit-lens and cross-conditioning analyses.

RESEARCH GRANTS

Thinking Machines Research Grant

Aug 2026

VESSL AI Compute Grant

May 2026

OpenAI Research Access Program

May 2024, Jul. 2024

PUBLICATIONS

- [1]. Samee Arif, et al., [One Word at a Time: Incremental Completion Decomposition Breaks LLM Safety](#). COLM 2026.
- [2]. Samee Arif, et al., [The Age of Curiosity Meets the Age of AI: Benchmarking Child Safety in LLMs](#). EMNLP 2026.
- [3]. Naihao Deng, Samee Arif, et al., [One Example Is Enough to Pass Fairness Benchmarks](#). EMNLP 2026.
- [4]. Samee Arif, et al., [With a Grain of SALT: Are LLMs Fair Across Social Dimensions?](#). TrustNLP at ACL 2026.
- [5]. Samee Arif, et al., [From Press to Pixels: Evolving Urdu Text Recognition](#). LREC 2026.
- [6]. Samee Arif, et al., [Kahaani: A Multimodal Co-Creative Storytelling System](#). SRW at EACL 2026.
- [7]. Samee Arif, et al., [The Fellowship of the LLMs: Multi-LLM Synthetic Preference Optimization](#). GEM2 at ACL 2025.
- [8]. Samee Arif, et al., [WER We Stand: Benchmarking Urdu ASR Models](#). COLING 2025.
- [9]. Samee Arif, et al., [Generalists vs. Specialists: Evaluating Large Language Models for Urdu](#). EMNLP 2024.
- [10]. Samee Arif, et al., [UQA: Corpus for Urdu Question Answering](#). LREC-COLING 2024.

TECHNICAL SKILLS

Languages: Python, C++, JavaScript, SQL, R

ML/AI: PyTorch, Hugging Face, Transformers, Pandas, vLLM, SFT, RLAI/RLVR, Activation Probing, CKA, Logit Lens

Systems/Tools: CUDA, Docker, Singularity, Git, Linux, Codex, Claude Code